

Business Performance Report

Data Analytics & Business Intelligence – Task 2

1. Introduction

This project focuses on analyzing the Superstore Sales dataset using SQL, Microsoft Excel, and Power BI. The objective was to combine business data, calculate key performance indicators (KPIs), identify sales trends, and develop an interactive dashboard to support business decision-making.

The analysis was carried out in three stages:

- SQL for data preparation and KPI calculations
 - Excel for pivot table analysis
 - Power BI for creating an interactive business dashboard
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2. KPI Overview

The following Key Performance Indicators (KPIs) were calculated during the analysis:

- **Total Sales:** ₹2.28 Million
- **Total Profit:** ₹283.08 Thousand
- **Total Orders:** 9,994
- **Profit Margin:** 12.42%

These KPIs provide an overview of the company's business performance and profitability.

3. Regional Performance Analysis

The Sales by Region analysis indicates that the **West** region generated the highest sales revenue, followed by the **East** region. The **South** region recorded the lowest sales among all regions.

This suggests that the company performs strongly in the West, while additional marketing efforts and customer engagement strategies could help improve sales performance in the South region.

4. Category Performance Analysis

Among all product categories, **Technology** generated the highest profit, followed by **Office Supplies**. **Furniture** produced the lowest profit despite contributing significantly to sales.

This indicates that Technology products offer better profitability, whereas Furniture products may require pricing optimization or better discount management.

5. Customer Analysis

The Consumer segment contributed the largest share of total sales, making it the company's most valuable customer segment.

The Top 10 Customers generated a significant portion of total revenue, highlighting the importance of maintaining strong customer relationships and implementing customer retention programs.

6. Sales Trend Analysis

Monthly sales analysis shows fluctuations throughout the year with noticeable peaks during certain months.

This indicates seasonal buying patterns, suggesting that promotional campaigns and inventory planning should be aligned with periods of higher customer demand.

7. Discount Impact Analysis

The analysis shows that higher discounts generally reduce overall profitability.

While discounts can increase sales volume, excessive discounting negatively affects profit margins. Therefore, discounts should be offered strategically to maximize both revenue and profitability.

8. Dashboard Summary

An interactive Power BI dashboard was developed to visualize business performance.

The dashboard includes:

- KPI Cards
- Sales by Region
- Profit by Category
- Monthly Sales Trend
- Sales by Customer Segment

- Top 10 Customers
- Interactive slicers for Region, Category, Segment, and Order Date

The dashboard enables users to filter data dynamically and analyze business performance from multiple perspectives.

9. Business Recommendations

Based on the analysis, the following recommendations are suggested:

1. Increase marketing activities in the South region to improve sales.
 2. Focus on expanding Technology product offerings due to their higher profitability.
 3. Optimize pricing and reduce excessive discounts on Furniture products.
 4. Strengthen customer loyalty programs for high-value customers.
 5. Use seasonal sales trends to improve inventory planning and promotional campaigns.
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10. Conclusion

This project demonstrates how SQL, Excel, and Power BI can be integrated to perform business intelligence analysis.

The dashboard provides valuable insights into sales performance, profitability, customer behavior, and regional trends, enabling management to make informed business decisions.

Overall, the project successfully transformed raw sales data into meaningful business insights through data analysis and visualization.